



SIMATS ENGINEERING



ASSESSMENT TOOL 2

CO3 AT2 Data Interpretation

Name: G. Sahitya

Reg no: 192424217

Slot: A

Course Code: DSA0404

Course Name: Fundamentals of Data Science

DOS: 17/08/2026

1. Sampling Distribution Reasoning

A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn.
(a) Compute the standard error of the sample mean. (b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal? (c) If n were increased to 256, how does the standard error change, and what principle explains this?

ANS:

Given:

Population mean (μ) = 500

Population standard deviation (σ) = 80

Sample size (n) = 64

(a) Compute the standard error of the sample mean

Formula:

$$SE = \sigma / \sqrt{n}$$

$$SE = 80 / \sqrt{64}$$

$$SE = 80 / 8$$

$$SE = 10$$

Answer: The standard error of the sample mean is 10.

(b) Probability that the sample mean exceeds 520

We need to find:

$$P(\bar{X} > 520)$$

First, calculate the z-score:

$$Z = (\bar{X} - \mu) / SE$$

$$Z = (520 - 500) / 10$$

$$Z = 20 / 10$$

$$Z = 2$$

From the standard normal distribution table:

$$P(Z > 2) = 0.0228$$

Therefore:

$$P(\bar{X} > 520) = 0.0228$$

Answer: The probability that the sample mean exceeds 520 is 0.0228 or approximately 2.28%.

(c) If n is increased to 256

New standard error:

$$SE = \sigma / \sqrt{n}$$

$$SE = 80 / \sqrt{256}$$

$$SE = 80 / 16$$

$$SE = 5$$

Answer: The standard error decreases from 10 to 5.

Principle:

The standard error decreases as the sample size increases. It is inversely proportional to the square root of the sample size:

$$SE = \sigma / \sqrt{n}$$

Thus, increasing the sample size makes the sample mean more precise.

2. Covariance Calculation and Sign Interpretation

Given paired data on advertising spend (X, ₹'000) and sales (Y, ₹ lakh):

X: 10, 20, 30, 40, 50

Y: 15, 18, 25, 30, 42

Calculate Cov(X,Y). Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

ANS:

Given data:

Advertising Spend (X): 10, 20, 30, 40, 50

Sales (Y): 15, 18, 25, 30, 42

Step 1: Calculate the mean of X

$$\bar{X} = (10 + 20 + 30 + 40 + 50) / 5$$

$$\bar{X} = 150 / 5$$

$$\bar{X} = 30$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (15 + 18 + 25 + 30 + 42) / 5$$

$$\bar{Y} = 130 / 5$$

$$\bar{Y} = 26$$

Step 3: Calculate covariance

Formula for sample covariance:

$$\text{Cov}(X, Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / (n - 1)$$

The required values are:

$$\Sigma[(X - \bar{X})(Y - \bar{Y})] = 660$$

Therefore:

$$\text{Cov}(X, Y) = 660 / (5 - 1)$$

$$\text{Cov}(X, Y) = 660 / 4$$

$$\text{Cov}(X, Y) = 165$$

Answer:

$$\text{Cov}(X, Y) = 165$$

Interpretation:

- The covariance is positive (165). This means that advertising spending and sales tend to increase together.
- The positive relationship is consistent with intuition because higher advertising spending is generally associated with higher sales.
- However, the magnitude of covariance alone does not tell us the strength of the relationship because covariance depends on the units and scales of the variables.

- Correlation would provide a better measure of the strength of the relationship because its value always lies between -1 and $+1$.

3. Correlation vs Causation Trap

A city reports that ice cream sales and drowning incidents have a correlation coefficient of $r = 0.85$. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

ANS:

Given:

Correlation coefficient (r) = 0.85

The correlation between ice cream sales and drowning incidents is strongly positive.

Statistical flaw:

The newspaper has incorrectly assumed that correlation implies causation.

A high correlation does not prove that ice cream sales cause drowning incidents.

Likely confounding variable:

The most likely confounding variable is temperature or summer weather.

For example:

Hot weather → Increased ice cream sales

Hot weather → More people swimming → Increased risk of drowning

Thus, both ice cream sales and drowning incidents may increase because of hot weather.

Additional evidence needed to claim causation:

Correlation alone is not sufficient to establish causation. Additional evidence would include:

1. Controlled experiments, where possible.
2. Randomized studies.
3. Controlling for confounding variables.
4. Longitudinal studies.
5. A plausible causal mechanism.
6. Consistent results across different studies and populations.

Conclusion:

The correlation coefficient of 0.85 indicates a strong positive association between ice cream sales and drowning incidents. However, it does not prove that ice cream sales cause drowning. The relationship is likely influenced by a confounding variable such as temperature or summer season.

4. Effect of Outliers on Correlation

A dataset of 9 points shows $r = 0.42$ (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

ANS:

Given:

Initial correlation coefficient:

$$r = 0.42$$

After removing one extreme outlier:

$$r = 0.89$$

Interpretation:

The correlation increased significantly from 0.42 to 0.89 after removing one extreme outlier.

This shows that Pearson's correlation is sensitive to extreme values or outliers. The outlier was strongly influencing the correlation and making the original relationship appear weaker.

Therefore, Pearson's correlation can be fragile when influential outliers are present.

Would Spearman's rank correlation be more robust?

Yes. Spearman's rank correlation would generally be more robust in this situation because it uses the ranks of the observations rather than their actual numerical values.

Pearson's Correlation:

- Uses actual numerical values.
- Measures linear relationships.
- Can be strongly affected by outliers.

Spearman's Correlation:

- Uses ranks of the data.
- Measures monotonic relationships.
- Is generally less sensitive to extreme values.

Conclusion:

The large change from $r = 0.42$ to $r = 0.89$ demonstrates the strong influence of the outlier. Spearman's rank correlation would generally be a more robust choice when outliers are present. However, the outlier should be investigated before removing it to determine whether it is a genuine observation or an error.

5. Data Preparation — Missing Values

A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents (not random).

(a) Classify this missing data mechanism (MCAR/MAR/MNAR). (b) Why would mean imputation introduce bias here? (c) Suggest a better strategy.

ANS:

Given:

Total number of rows = 200

Missing values in Income column = 15%

The missing values are more common among unemployed respondents.

(a) Classify the missing data mechanism

The missing data mechanism is MAR (Missing At Random).

Reason:

The missingness of Income is related to employment status, which is an observed variable in the dataset.

Answer: MAR (Missing At Random)

(b) Why would mean imputation introduce bias?

Mean imputation replaces all missing Income values with the overall mean income.

However, the missing values are more common among unemployed respondents. Their income is likely to be systematically different from the income of employed respondents.

Therefore, using the overall mean may:

1. Overestimate or underestimate the income of unemployed respondents.
2. Distort the income distribution.
3. Reduce the variance of the dataset.
4. Affect the relationship between Income and other variables.
5. Produce biased analysis results.

Thus, simple mean imputation is not appropriate for this dataset.

(c) Suggest a better strategy

A better strategy is to use an imputation method that considers employment status and other relevant variables.

Suitable methods include:

1. Group-based imputation based on employment status.
2. Regression-based or model-based imputation.
3. Multiple imputation.

For example, missing Income values can be estimated using variables such as employment status, age, education, occupation, and other income-related variables.

Multiple imputation is generally a good choice for MAR data because it accounts for uncertainty in the estimated values.

Conclusion:

Since the missing Income values are related to employment status, simple mean imputation should be avoided. A group-based, regression-based, or multiple-imputation method would provide more reliable and less biased results.